

Siddarth Shinde

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EDUCATION

Northwestern University	Starting Sep. 2026; Ph.D. , Computer Engineering
North Carolina State University	May 2026; B.S. , Computer Science; GPA: 3.87
	Minors , Mathematics and Philosophy

RESEARCH EXPERIENCE

My research areas are quantum computing systems and architectures, quantum computer security, and quantum algorithms.

Northwestern University	06/2025 – 08/2025	Undergraduate Research Assistant
Department of Electrical and Computer Engineering (Evanston, IL, USA)		Supervisor: Prof. Jakub Szefer
North Carolina State University	05/2024 – 08/2024	Undergraduate Research Assistant
Department of Computer Science (Raleigh, NC, USA)		Supervisor: Prof. Chin Ho Lee

INDUSTRY EXPERIENCE

apexanalytix Data Science (Greensboro, NC, USA)	Summer 2026	Quantum Computing Intern
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TEACHING EXPERIENCE

North Carolina State University	08/2024 12/2024	– Undergraduate Teaching Assistant
Department of Computer Science (Raleigh, NC, USA)		Supervisor: Prof. Abida Haque Course: CSC 333
North Carolina State University	01/2024 05/2024	– Undergraduate Teaching Assistant
Department of Computer Science (Raleigh, NC, USA)		Supervisor: Prof. Suzanne Balik Course: CSC 116
North Carolina State University	08/2023 12/2023	– Undergraduate Teaching Assistant
Department of Computer Science (Raleigh, NC, USA)		Supervisor: Prof. Thomas Price Course: CSC 110

PUBLICATIONS AND PAPERS

- **Quantum Fidelity-per-Cost: A Metric for Evaluation of Quantum Computing Systems.** Siddarth Shinde and Jakub Szefer. *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026. Accepted. arXiv:2607.28572.

AWARDS AND FELLOWSHIPS

- Royal E. Cabell Fellowship, Northwestern University, 2026.

SELECTED PROJECTS AND TECHNICAL SKILLS

- **OceanVoice** (React, TypeScript, Python, Flask), 2026. Built a voice-first multimodal interface for OceanAI with transcript-driven execution, clarification, confirmation, and LLM tool-calling workflows.
- **Grover’s Search Lower Bound**, 2025. Wrote a formal [technical report](#) proving the $\Omega(\sqrt{N})$ quantum query lower bound for unstructured search using the BBBV hybrid argument.
- **Technical skills:** Python, Java, JavaScript/TypeScript, C, SQL, x86 Assembly; Qiskit, AWS Braket, IBM Quantum; Git, Docker, Linux, Flask, React, Spring, Hibernate, JUnit, and ROS.